Bayesian Model - NYC 311 Resolution Time

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2026-04-02

```r
library(here)
library(brms)
library(tidybayes)
# load preprocessed sample directly - full CSV is too large to reload
df_model <- readRDS(here("results", "df_model.rds"))
head(df_model)
```

```
##        Complaint.Type   Borough        Created.Date         Closed.Date
## 1    Blocked Driveway    QUEENS 2019-09-22 16:22:20 2019-09-22 22:18:03
## 2      Taxi Complaint MANHATTAN 2019-09-23 16:27:35 2019-11-26 16:15:00
## 3     Illegal Parking  BROOKLYN 2019-09-23 19:41:14 2019-09-23 23:45:43
## 4    Blocked Driveway    QUEENS 2019-09-23 20:06:33 2019-09-25 00:02:52
## 5 Urinating in Public MANHATTAN 2019-09-23 19:41:42 2019-09-23 21:34:54
## 6     Illegal Parking  BROOKLYN 2019-09-23 21:00:38 2019-09-23 23:49:08
##   resolution_hours hour day_of_week hour_scaled day_of_week_scaled
## 1         5.928611   16           7   0.2447453          1.3614180
## 2      1383.395556   16           1   0.2447453         -0.7530197
## 3         4.074722   19           1   0.7566497         -0.7530197
## 4        27.938611   20           1   0.9272845         -0.7530197
## 5         1.886667   19           1   0.7566497         -0.7530197
## 6         2.808333   21           1   1.0979193         -0.7530197
##   log_resolution
## 1      1.7797900
## 2      7.2322963
## 3      1.4048026
## 4      3.3300096
## 5      0.6348116
## 6      1.0325912
```

```r
default_prior(log_resolution ~ Borough + hour_scaled + day_of_week_scaled + (1|Complaint.Type),
              data = df_model,
              family = gaussian())
```

```
##                    prior     class                coef          group resp dpar
##                   (flat)         b
##                   (flat)         b     BoroughBROOKLYN
##                   (flat)         b    BoroughMANHATTAN
##                   (flat)         b       BoroughQUEENS
##                   (flat)         b BoroughSTATENISLAND
##                   (flat)         b  BoroughUnspecified
##                   (flat)         b  day_of_week_scaled
```

```
##                  (flat)         b         hour_scaled
##  student_t(3, 2.2, 2.5) Intercept
##    student_t(3, 0, 2.5)        sd
##    student_t(3, 0, 2.5)        sd                     Complaint.Type
##    student_t(3, 0, 2.5)        sd           Intercept Complaint.Type
##    student_t(3, 0, 2.5)     sigma
##  nlpar lb ub tag       source
##                        default
##                  (vectorized)
##                  (vectorized)
##                  (vectorized)
##                  (vectorized)
##                  (vectorized)
##                  (vectorized)
##                  (vectorized)
##                        default
##         0              default
##         0        (vectorized)
##         0        (vectorized)
##         0              default
```

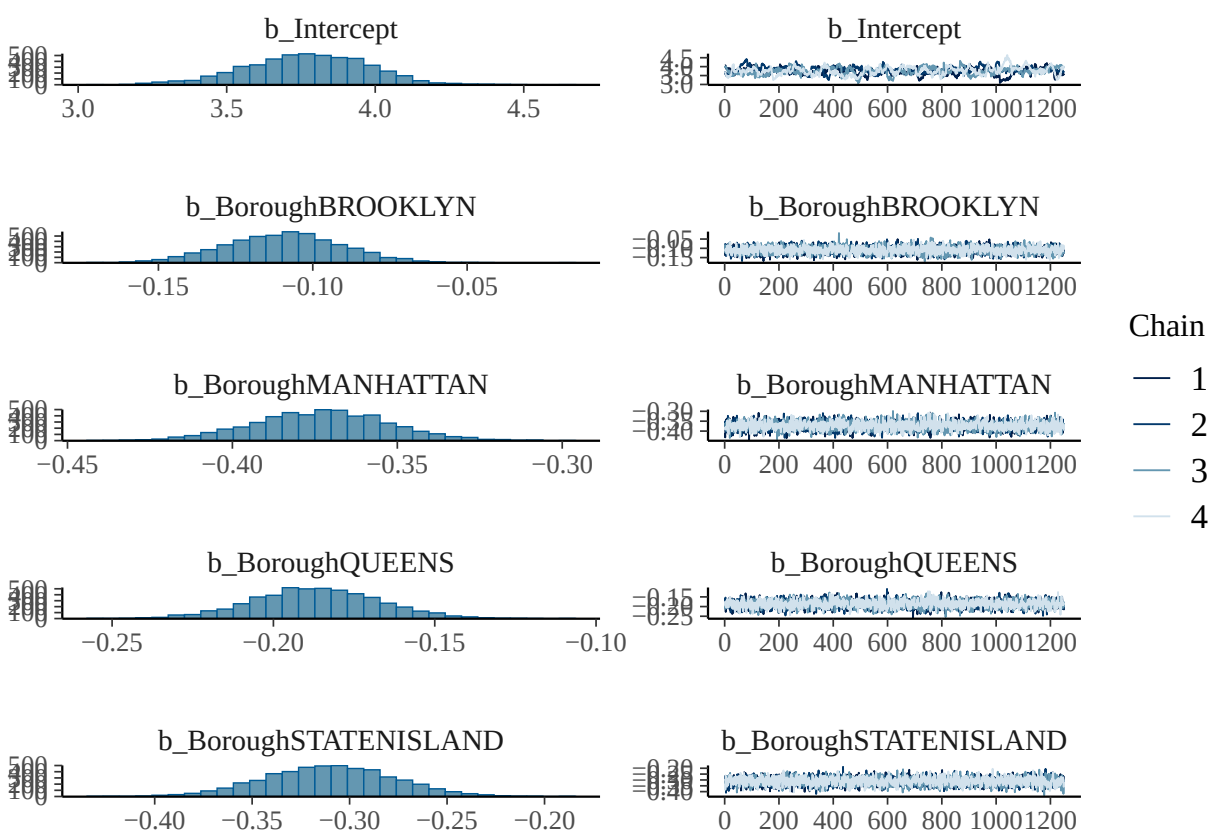
```r
model_path <- here("results", "nyc_brm.rds")

if (file.exists(model_path)) {
  nyc_brm <- readRDS(model_path)
} else {
  nyc_brm <- brm(
    log_resolution ~ Borough + hour_scaled + day_of_week_scaled + (1|Complaint.Type),
    family  = gaussian(),
    data    = df_model,
    chains  = 4,
    cores   = 4,
    iter    = 3000,
    warmup  = 500,
    thin    = 2,
    seed    = 42
  )
  saveRDS(nyc_brm, model_path)
}
```

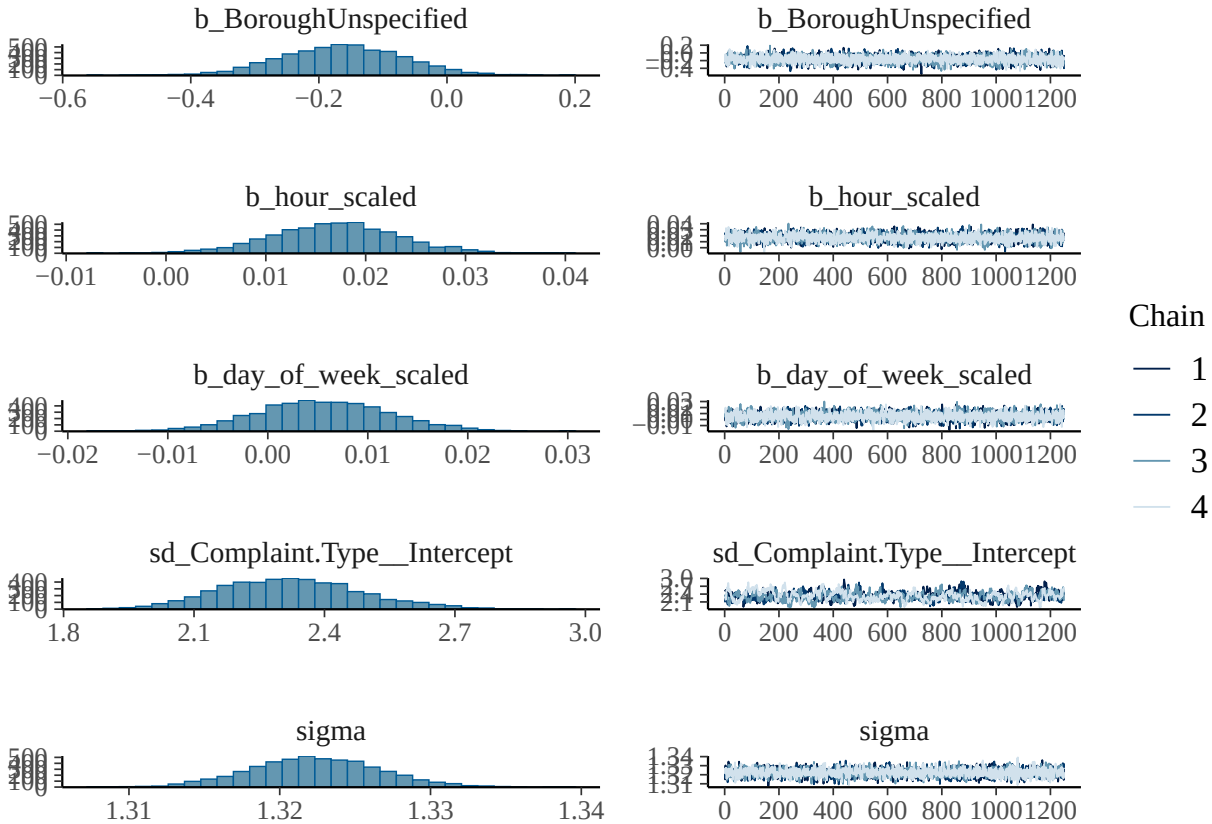
```r
summary(nyc_brm)
```

```
##  Family: gaussian
##   Links: mu = identity
## Formula: log_resolution ~ Borough + hour_scaled + day_of_week_scaled + (1 | Complaint.Type)
##    Data: df_model (Number of observations: 50000)
##   Draws: 4 chains, each with iter = 3000; warmup = 500; thin = 2;
##          total post-warmup draws = 5000
##
## Multilevel Hyperparameters:
## ~Complaint.Type (Number of levels: 107)
##               Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sd(Intercept)     2.32      0.16     2.03     2.66 1.01      301      581
```

```
##
## Regression Coefficients:
##                      Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## Intercept                3.77      0.21     3.34     4.15 1.03      185      273
## BoroughBROOKLYN         -0.11      0.02    -0.15    -0.07 1.00     3527     4194
## BoroughMANHATTAN        -0.37      0.02    -0.41    -0.33 1.00     3619     4671
## BoroughQUEENS           -0.19      0.02    -0.23    -0.15 1.00     3198     3963
## BoroughSTATENISLAND     -0.31      0.03    -0.37    -0.25 1.00     3868     4151
## BoroughUnspecified      -0.16      0.09    -0.35     0.02 1.00     4802     4549
## hour_scaled              0.02      0.01     0.00     0.03 1.00     4256     4138
## day_of_week_scaled       0.01      0.01    -0.01     0.02 1.00     4046     4051
##
## Further Distributional Parameters:
##       Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sigma     1.32      0.00     1.31     1.33 1.00     4324     3814
##
## Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
## and Tail_ESS are effective sample size measures, and Rhat is the potential
## scale reduction factor on split chains (at convergence, Rhat = 1).
```

```r
# Check convergence and posterior distributions
plot(nyc_brm)
```



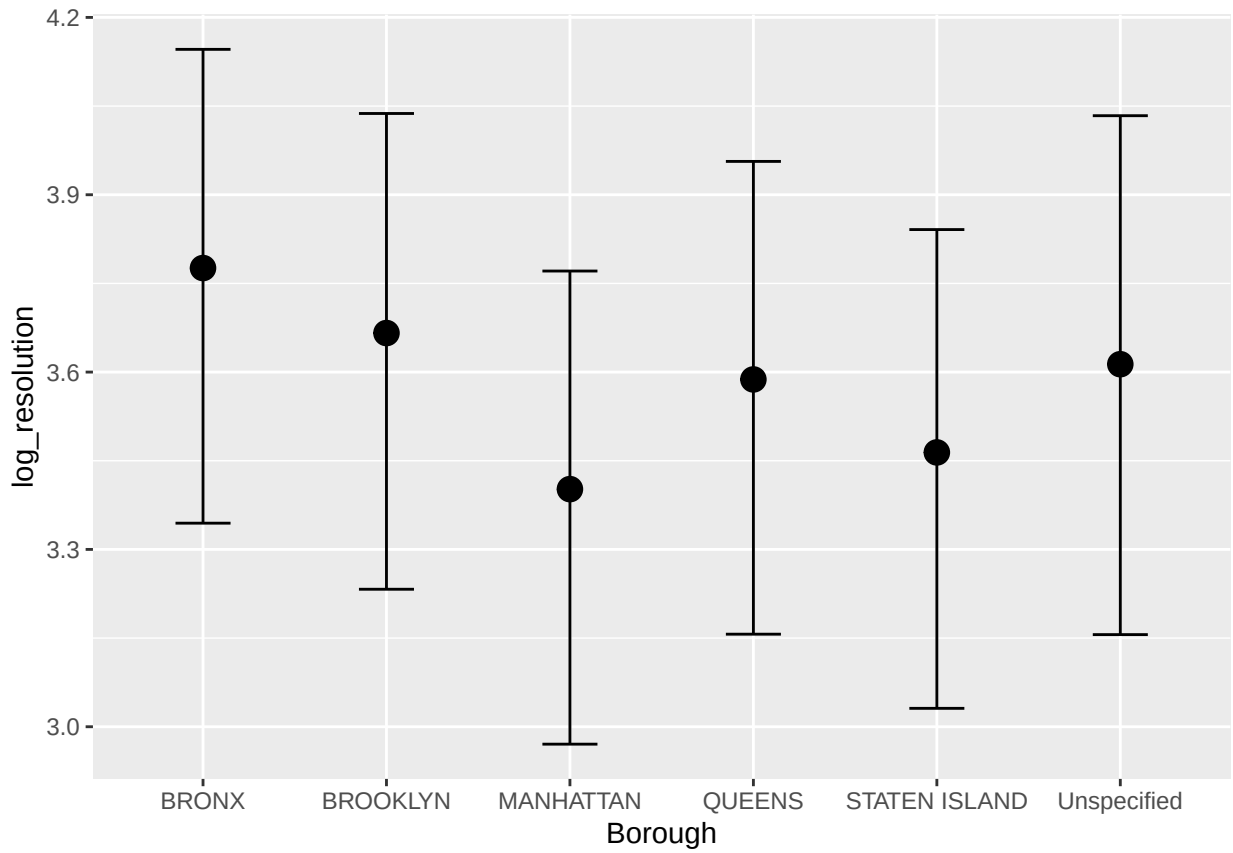



```r
# Bayesian R^2
bayes_R2(nyc_brm)
```

```
##     Estimate   Est.Error      Q2.5     Q97.5
## R2 0.6734379 0.001372822 0.6707492 0.6760867
```

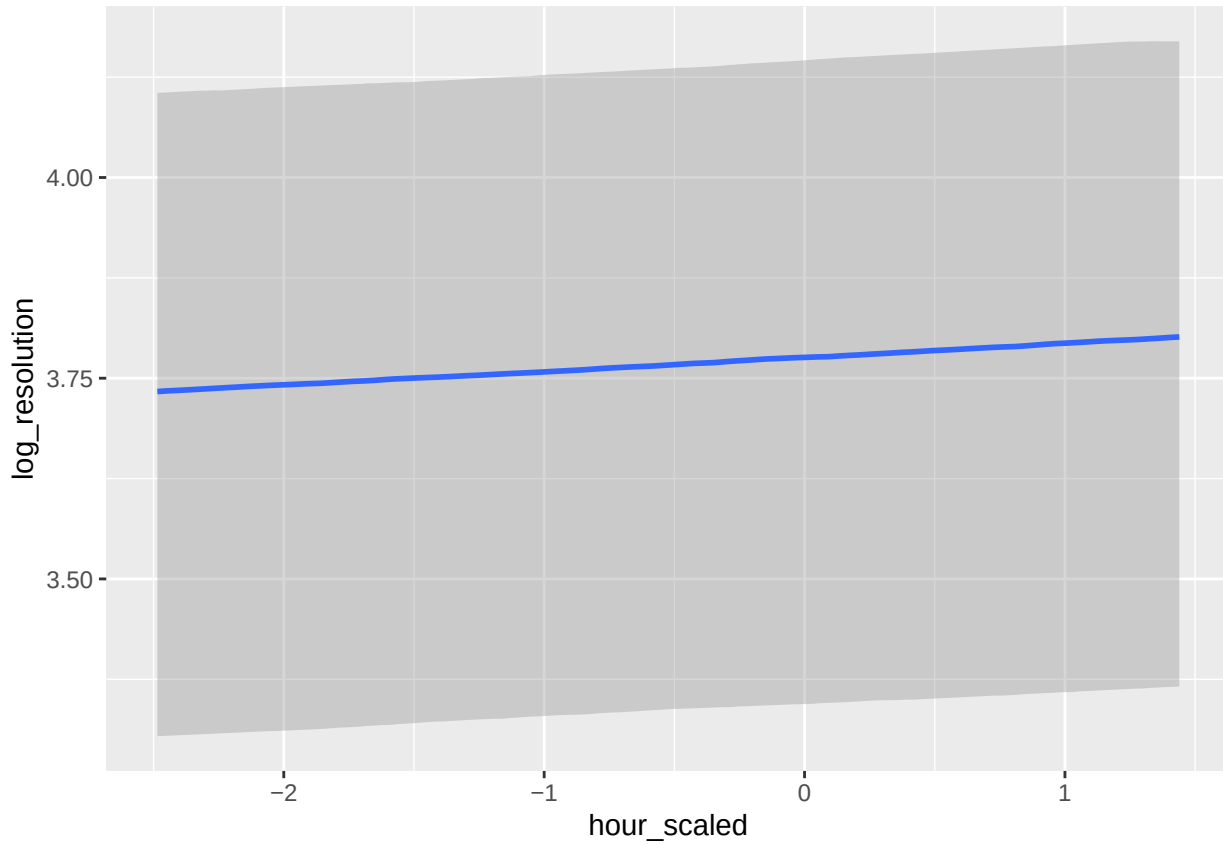
```r
# How does each feature affect log_resolution?
plot(conditional_effects(nyc_brm, "Borough"))
```

```
## Ignoring unknown labels:
## * fill : "NA"
## * colour : "NA"
## Ignoring unknown labels:
## * fill : "NA"
## * colour : "NA"
```

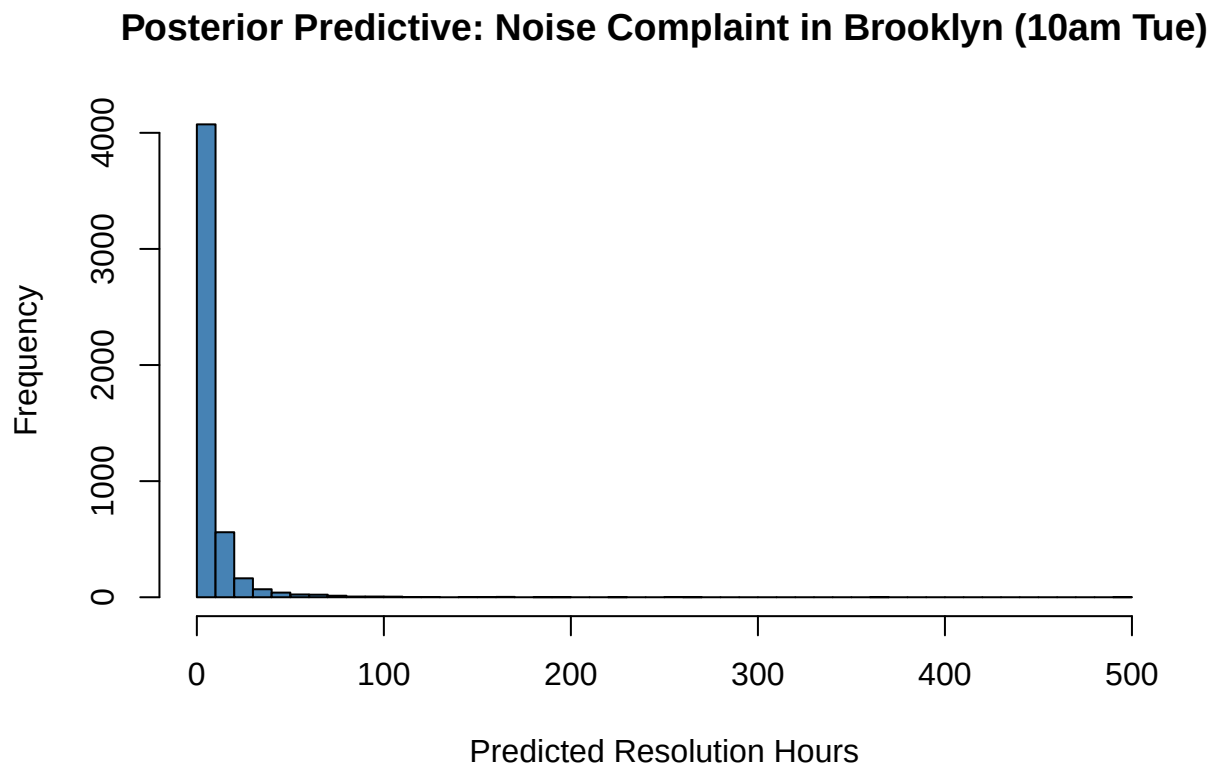



```r
plot(conditional_effects(nyc_brm, "hour_scaled"))
```

```
## Ignoring unknown labels:
## * fill : "NA"
## * colour : "NA"
## Ignoring unknown labels:
## * fill : "NA"
## * colour : "NA"
```




```r
# Posterior predictive distribution for a specific example
# "A noise complaint in Brooklyn at 10am on a Tuesday"
# hour_scaled: (10 - mean(hour)) / sd(hour)
# day_of_week_scaled: Tuesday = 2
hour_mean <- mean(df_model$hour)
hour_sd <- sd(df_model$hour)
dow_mean <- mean(df_model$day_of_week)
dow_sd <- sd(df_model$day_of_week)

example_request <- data.frame(
  Borough = factor("BROOKLYN", levels = levels(df_model$Borough)),
  hour_scaled = (10 - hour_mean) / hour_sd,
  day_of_week_scaled = (2 - dow_mean) / dow_sd,
  Complaint.Type = factor("Noise - Residential", levels = levels(df_model$Complaint.Type))
)

post_preds <- posterior_predict(nyc_brm, newdata = example_request, allow_new_levels=TRUE)
hist(exp(post_preds),
     main = "Posterior Predictive: Noise Complaint in Brooklyn (10am Tue)",
     xlab = "Predicted Resolution Hours",
     col = "steelblue", breaks = 50)
```




```r
cat("Posterior mean resolution:", round(mean(exp(post_preds)), 1), "hours\n")
```

```
## Posterior mean resolution: 7.5 hours
```

```r
# Which complaint types have the widest uncertainty?
library(dplyr)
library(tidybayes)
nyc_brm %>%
  spread_draws(r_Complaint.Type[Complaint.Type, ]) %>%
  group_by(Complaint.Type) %>%
  summarise(
    mean = mean(r_Complaint.Type),
    sd = sd(r_Complaint.Type),
    q05 = quantile(r_Complaint.Type, 0.05),
    q95 = quantile(r_Complaint.Type, 0.95)
  ) %>%
  arrange(desc(sd)) %>%
  print(n = 20)
```

```
## # A tibble: 107 x 5
##    Complaint.Type                      mean    sd    q05    q95
##    <chr>                              <dbl> <dbl>  <dbl>  <dbl>
##  1 Cranes.and.Derricks               -0.769  0.753 -1.98   0.475
##  2 BEST/Site.Safety                   0.241  0.751 -0.985  1.50
##  3 Illegal.Fireworks                 -1.57   0.671 -2.68  -0.471
```

```
##  4 Bike.Rack.Condition                  2.35   0.663  1.25   3.45
##  5 Ferry.Inquiry                        0.279  0.608 -0.727  1.26
##  6 Executive.Inspections                3.42   0.598  2.40   4.39
##  7 Public.Payphone.Complaint            2.41   0.541  1.52   3.30
##  8 Plant                               -2.98   0.527 -3.85  -2.10
##  9 Boilers                              0.571  0.526 -0.301  1.43
## 10 Investigations.and.Discipline.(IAD) -0.0429 0.484 -0.837  0.752
## 11 Graffiti                            -3.33   0.460 -4.10  -2.57
## 12 Noise.-.House.of.Worship            -3.74   0.419 -4.41  -3.04
## 13 DEP.Street.Condition                -2.23   0.415 -2.92  -1.54
## 14 Mold                                1.15   0.408  0.486  1.84
## 15 Bus.Stop.Shelter.Placement          0.767  0.402  0.114  1.42
## 16 ELEVATOR                            0.891  0.396  0.247  1.55
## 17 Industrial.Waste                   -2.08   0.388 -2.72  -1.43
## 18 Taxi.Compliment                     3.20   0.381  2.58   3.84
## 19 Drinking.Water                      3.01   0.380  2.39   3.64
## 20 Other.Enforcement                   1.08   0.342  0.523  1.65
## # i 87 more rows
```